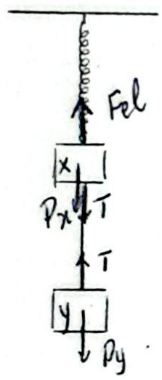
Colégio
Agostiniano Mendel

Componente Curricular: Física – Dinâmica
Professor(a): Rosana C. **Série:** 2ª **Data:** Fevereiro/2026
Conteúdo: Força Elástica - Exercícios - Resolução

1- Diagrama ⇒ a) $F = k \cdot \Delta x \Rightarrow 100 = k \cdot 0,2 \Rightarrow k = \frac{100}{0,2} \Rightarrow \underline{k = 500\ N/m}$

b) $F = k \cdot \Delta x \Rightarrow F = 500 \cdot 0,05 \Rightarrow \underline{F = 25\ N}$

2-



$Fel = 15N$

$m_x = m$

$m_y = 2m$

Repouso $F_R = 0$

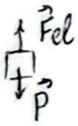
. $y \Rightarrow T = P_y \Rightarrow T = 20m$

. $X \Rightarrow P_x + T = Fel \Rightarrow 10m + T = 15 \Rightarrow 10m + 20m = 15 \Rightarrow 30m = 15$

$\underline{m = 0,5 kg}$

. $T = 20 \cdot 0,5 \Rightarrow T = 10N$

3.



Repouso ⇒ $F_R = 0$

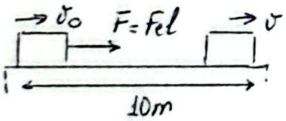
$Fel = P \Rightarrow k \cdot x = P \Rightarrow k \cdot 0,2 = 16$

$k = \frac{16}{0,2} \Rightarrow \boxed{k = 80\ N/m}$

$* \ x = L_F - L_0 \Rightarrow x = 1,4 - 1,2$

$\cdot \ x = 0,2m$

4.



$* \ v^2 = v_0^2 + 2 \cdot a \cdot \Delta S$

$6^2 = 4^2 + 2 \cdot a \cdot 10$

$36 - 16 = 20 \cdot a \Rightarrow 20 = 20 \cdot a$

$a = 1\ m/s^2$

$* \ Fel = m \cdot a$

$k \cdot x = m \cdot a$

$160 \cdot x = 4 \cdot 1 \Rightarrow \boxed{x = \frac{1}{40} m}$

$x = 0,025m = 2,5cm$

$\cdot \ x = l - l_0$

$2,5 = l - 14$

$\boxed{l = 16,5\ cm}$

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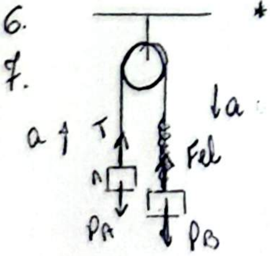
5. $F = k \cdot x \Rightarrow 8 = k \cdot 0,2 \Rightarrow k = 40 N/m$

6.

7.



$* \ T = Fel$ (fio ideal)

$* \ A \Rightarrow T - P_A = m_A \cdot a$

$T - 10 = 1 \cdot 3$

$T = 13N \therefore \underline{Fel = 13N}$

$* \ Fel = k \cdot x$

$13 = 26 \cdot x \Rightarrow x = \frac{13}{26} \Rightarrow x = 0,5m$

$* \ x = L - L_0 \Rightarrow 0,5 = L - 1 \Rightarrow \boxed{L = 1,5 m}$

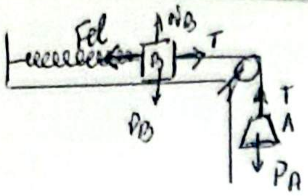
$* \ B \Rightarrow P_B - Fel = m_B \cdot a$

$10 m_B - 13 = 3 m_B$

$7 m_B = 13$

$m_B = \frac{13}{7} \Rightarrow \boxed{m_B = 1,86\ kg}$

8.



Repouso ($F_R = 0$)

$x = 5cm = 0{,}05m$

A ⇒ $T = P_A$

B ⇒ $F_{el} = T$

∴ $F_{el} = P_A$

$k.x = P_A$

$k.0{,}05 = 10$

$k = \frac{10}{0{,}05}$ ⇒ $k = 200 N/m$

9.

· $m_1 = m_2 = m$

$P_1 = P_2 = 100N$

* $m = 10kg$

$F_R = m.a$ ⇒ 1 ⇒ $F + P_1 - F_{el} = m.a$

2 ⇒ $F_{el} + P_2 = m.a$

$F + P_1 + P_2 = 2m.a$

$200 + 100 + 100 = 20.a$

$400 = 20.a$

$a = 2 m/s^2$

* 2 ⇒ $F_{el} + 10 = 1.2$

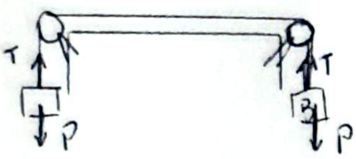
$F_{el} = 10N$

* $F_{el} = k.x$

$10 = 1000.x$ ⇒ $x = 0{,}01m$

$x = 10cm$

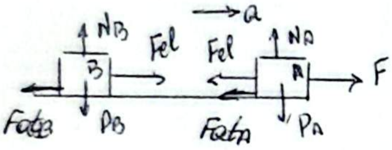
10.



Dinamômetro ⇒ Indica T

A e B ⇒ $F_R = 0$ (repouso) $T = P$ ⇒ $T = 100N$

11.



* $F_{at} = \mu.N$

· $F_{at_A} = 0{,}4.60 = 24N$

· $F_{at_B} = 0{,}4.40 = 16N$

$F_R = m.a$ ⇒ B ⇒ $F_{el} - F_{at_B} = m_A.a$

A ⇒ $F - F_{el} - F_{at_A} = m_A.a$

$F - F_{at_A} - F_{at_B} = (m_A + m_B).a$

$60 - 24 - 16 = 10.a$

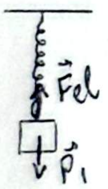
$20 = 10.a$ ⇒ $a = 2 m/s^2$

* B ⇒ $F_{el} - 16 = 4.2$ ⇒ $F_{el} = 16 + 8$

$F_{el} = 24N$

* $F_{el} = k.x$ ⇒ $24 = 800.x$ ⇒ $x = 0{,}03m$ ⇒ $\boxed{x = 3cm}$

12.

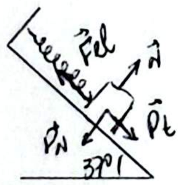


Repouso ⇒ $F_R = 0$

$F_{el} = P_1$

$k.x_1 = P_1$

$k.8 = 12$ ⇒ $\boxed{k = 1{,}5 N/cm}$



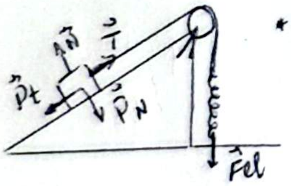
Repouso ⇒ $F_R = 0$

$F_{el} = P_t$

$k.x = P_2.sen37°$

$1{,}5.x = 10.0{,}6$ ⇒ $1{,}5x = 6$ ⇒ $x = 4cm$

13.



* $F_{el} = T$ (fio ideal)

Repouso ⇒ $F_R = 0$

$P_t = F_{el}$

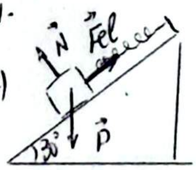
$P.sen30° = k.x$

$30.0{,}5 = 0{,}3.x$

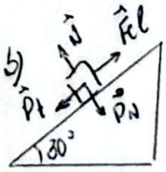
$\boxed{x = 50m}$

14.

a)



b)



Repouso ($F_R = 0$)

$F_{el} = P_t$

$k.x = P_t$

$100.x = 50.0{,}5$

$x = 0{,}025m$

$\boxed{x = 25cm}$